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1.

(1) Prove that $\int_{0}^{1} x^{-x} \, \mathrm{d}=\sum_{n\geqslant 1}^{}{}n^{-n}$.
(2) Prove that $\int_{[0,1]^{2}}^{} (xy)^{-xy} \, \mathrm{d}x\mathrm{d}y=\sum_{}^{}{n^{-n}}$.
(3) Calculate $\int_{0}^{1} x^{x} \, \mathrm{d}x$ and $\int_{[0,1]^{2}}^{} (xy)^{xy} \, \mathrm{d}x\mathrm{d}y$.
(4) Calculate $\int_{[0,1]^{n}}^{} \left(\prod_{}^{}{x_{i}} \right)^{-\prod_{}^{}{x_{i}}} \, \mathrm{d}x_{1}\cdots \mathrm{d}x_{n}$.
Proof: (1) Note that $x^{-x}=\exp(-x\log x)=\sum_{n=0}^{\infty}{\frac{1}{n!}(-x\log x)^{n}}$. Since $(-x\log x)>0$, monotone convergence gives $\int_{0}^{1} x^{-x} \, \mathrm{d}x=\sum_{n\geqslant 0}^{}{\int_{0}^{1} \frac{1}{n!}(-x\log x)^{n} \, \mathrm{d}x}$. Note that $\int_{0}^{1} (x\log x)^{n} \, \mathrm{d}x=\int_{-\infty}^{0} e^{t(n+1)}t^{n} \, \mathrm{d}t=\frac{(-1)^{n}}{(n+1)^{n+1}}\Gamma(n+1)$, hence $\int_{0}^{1} x^{-x} \, \mathrm{d}x=\sum_{n\geqslant 1}^{}{n^{-n}}$.
(2) Analogously,

$$\begin{aligned}\int_{[0,1]^2}(xy\log xy)^n\,\mathrm{d}x\mathrm{d}y&=(-1)^n\int_{(0,\infty)^2}e^{-(s+t)(n+1)}(s+t)^n\,\mathrm{d}s\mathrm{d}t\\&=\frac{(-1)^n}{(n+1)^{n+2}}\Gamma(n+2).\end{aligned}$$

Hence $\int_{[0,1]^{2}}^{} (xy)^{-xy} \, \mathrm{d}x\mathrm{d}y=\sum_{}^{}{n^{-n}}$.
(3) $\int_{0}^{1} x^{x} \, \mathrm{d}x=\int_{[0,1]^{2}}^{}(xy)^{xy} \, \mathrm{d}x\mathrm{d}y=\sum_{n\geqslant 1}^{}{(-1)^{n+1}n^{-n}}$.
(4) Analogously,

$$\int_{[0,1]^n}\left(\prod x_i\right)^{-\prod x_i}\mathrm{d}x_1\cdots\mathrm{d}x_n=\sum_{k\geqslant 1}\binom{n+k-2}{n-1}k^{1-k-n}.$$

17.1

17.1.1

Suppose $\sigma:[\alpha,\beta]\to \mathbb{R}^{n}$ is BV and $\phi:[a,b]\to[\alpha,\beta]$ is a homeomorphism, prove that $\sigma\circ\phi$ and σ have the same length.
Proof: For any partition $\alpha=x_0<\cdots<x_n=\beta$, let $y_i=\phi^{-1}(x_i)$, then $L(\sigma;P)=\sum|\sigma(x_i)-\sigma(x_{i+1})|$ $=L(\sigma\circ\phi,\phi^{-1}(P))\leqslant L(\sigma\circ\phi)$. Hence $L(\sigma)\leqslant L(\sigma\circ\phi)$ and likewise $L(\sigma\circ\phi)\leqslant L(\sigma)$. Therefore they have the same length.

17.1.2

Suppose σ is a continuous BV curve, prove that $L(\sigma)=\lim_{\|P\|\to 0}L(\sigma;P)$.
Proof: For each $\varepsilon>0$, take a partition Q such that $L(\sigma;Q)>L(\sigma)-\varepsilon$. If $\|P\|<\delta$, let $\omega(\delta)$ $=\sup_{|x-y|<3\delta}|\sigma(x)-\sigma(y)|$, then for $|x-z|<2\delta$, we have $|\sigma(x)-\sigma(y)|+|\sigma(y)-\sigma(z)|-|\sigma(x)-\sigma(z)|$ $\leqslant 3\omega(\delta)$ (where $x<y<z$). Hence $L(\sigma;P\cup Q)\leqslant L(\sigma;P)+3m\omega(\delta)$ where m is the number of point in Q. Since $L(\sigma;P\cup Q)\geqslant L(\sigma;Q)>L(\sigma)-\varepsilon$, let $\delta\to 0$ we have $\liminf L(\sigma;P)\geqslant L(\sigma)-\varepsilon$, so let $\varepsilon\to 0$ we have $L(\sigma)=\lim_{\|P\|\to 0}L(\sigma;P)$.

17.1.4

(1) $\int_\sigma y^2\,\mathrm{d}s$ where $\sigma:x(t)=a(t-\sin t),y(t)=a(1-\cos t),0\leqslant t\leqslant 2\pi$:
Then

$$\int_\sigma y^2\,\mathrm{d}s=\int_0^{2\pi}\sqrt{2}a^3(1-\cos t)^{5/2}\,\mathrm{d}t=\frac{256}{15}a^3.$$

(3) $\int_\sigma\sqrt{x^2+y^2}\,\mathrm{d}s$ where $\sigma:x^2+y^2=ax$.
Consider parametrization $x=\frac{a}{1+t^2},y=\frac{ta}{1+t^2}$ with $t\in \mathbb{R}$, then $\int_{\sigma}^{} \sqrt{ x^{2}+y^{2} } \, \mathrm{d}s=\int_{\mathbb{R}}^{} a^{2}(1+t^{2})^{-3/2} \, \mathrm{d}t=2a^{2}$.
(5) $\int_\sigma x^2\,\mathrm{d}s$ where $\sigma:x^2+y^2+z^2=a^2,x+y+z=0$.
Note that $\int_\sigma x^2\,\mathrm{d}s=\int_\sigma y^2\,\mathrm{d}s=\int_\sigma z^2\,\mathrm{d}s$, so $\int_\sigma x^2\,\mathrm{d}s=\frac{a^2}{3}\int_\sigma 1\,\mathrm{d}s=\frac{2\pi a^3}{3}$.

17.1.6

Suppose some substance is evenly distributed on $\sigma(t)=(a(t-\sin t),a(1-\cos t)),0\leqslant t\leqslant\pi$. Determine the center of mass.
Solution: The center of mass is at $\left(\frac{4}{3}a,\frac{4}{3}a\right)$.

17.1.8

Suppose $\sigma:[\alpha,\beta]\to \mathbb{R}^{n}$ is continuous and injective. If $\sigma\in\mathrm{BV}([\alpha,\beta])$, prove that

$$\lim_{m\to\infty}\sup\{|t''-t'|:L(\sigma|_{[t',t'']})\leqslant m^{-1}\}=0.$$

Proof: Let $f(t)=L(\sigma|_{[\alpha,t]})$, then f is continuous and $L(\sigma|_{[t,t']})=f(t')-f(t)$. Hence the desired equality comes from uniform continuity of f^{-1}.

17.1.9

Suppose $f\in C^1[\alpha,\beta]$, prove that $\bigvee_\alpha^\beta(f)=\int_\alpha^\beta|f'(t)|\,\mathrm{d}t$.
Proof: Given a partition $\alpha=x_0<\cdots<x_n=\beta$,

$$V(f;P)=\sum|f(x_i)-f(x_{i+1})|=\sum|f'(\eta_i)|\cdot(x_{i+1}-x_i).$$

Since $f\in C^1[\alpha,\beta]$, the right side converges to $\int_\alpha^\beta|f'(t)|\,\mathrm{d}t$ as $\|P\|\to 0$, and a finer partition always has larger variation, hence $\int_\alpha^\beta|f'(t)|\,\mathrm{d}t=\sup V(f;P)=\bigvee_\alpha^\beta(f)$.

17.2

17.2.1

(1) $\int_\sigma x\,\mathrm{d}x + y\mathrm{d}y + z\mathrm{d}z$ where σ is the segment from $(1,1,1)$ to $(2,3,4)$.
Solution: $\sigma(t) = (1,1,1) + t(1,2,3)$, so $\int_\sigma x\,\mathrm{d}x = \int_0^1 6 + 14t\,\mathrm{d}t = 13$.

17.2.2

Calculate $\int_\sigma (y-z)\,\mathrm{d}x + (z-x)\mathrm{d}y + (x-y)\mathrm{d}z$ where $\sigma: x^2+y^2=1, x+z=1$.
Solution: $\sigma(t) = (\cos t, -\sin t, 1-\cos t)$ for $t \in [0, 2\pi]$, so the integral is 4π.

17.2.3

Calculate $\int_\sigma (y^2-z^2)\,\mathrm{d}x + (z^2-x^2)\mathrm{d}y + (x^2-y^2)\mathrm{d}z$ where σ is the boundary of $x^2+y^2+z^2=1, x \geqslant 0, y \geqslant 0, z \geqslant 0$ of counter-clockwise orientation.
Solution: Let $F = (y^2-z^2, z^2-x^2, x^2-y^2)$, then $\nabla \times F = -2(y+z, z+x, x+y)$ so Stokes theorem implies $\int_\sigma F\,\mathrm{d}x = \int_{\mathrm{int}(\sigma)} -4(xy+yz+zx)\,\mathrm{d}S = -12\int_{\mathrm{int}(\sigma)} xy\,\mathrm{d}S$. By change of variables $x = \sin\theta\sin\psi, y = \cos\theta\sin\psi, dS = \sin\psi d\psi d\theta$, so the integral is $-12\int_{[0,\pi/2]^2} \sin^3\psi\sin\theta\cos\theta\,\mathrm{d}\psi\mathrm{d}\theta = -4$.

17.2.4

Calculate $\int_\sigma \frac{y\mathrm{d}x - x\mathrm{d}y}{x^2+y^2}$ where $\sigma: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ counter-clockwise.
Solution: $\sigma(t) = (a\cos t, b\sin t)$ for $t \in [0, 2\pi]$, so the integral is

$$\int_0^{2\pi} \frac{-a^2\cos^2 t - b^2\sin^2 t}{a^2\cos^2 t + b^2\sin^2 t}\mathrm{d}t = -2\pi.$$

(Or notice that this is just the winding number.)

17.2.7

Suppose P, Q is continuous on the BV curve σ, prove that

$$\left|\int_\sigma P(x,y)\,\mathrm{d}x + Q(x,y)\mathrm{d}y\right| \leqslant L(\sigma)M,$$

where $M = \sup_\sigma \sqrt{P^2+Q^2}$.
Proof: Assume $\sigma=(x(t),y(t)):[a,b]\to \mathbb{R}^{2}$ and take any partition $a = t_0 < \cdots < t_n = b$, then

$$\left|\sum (P(\sigma(t_k))x(t_k) + Q(\sigma(t_k))y(t_k))(t_k - t_{k-1})\right| \leqslant M\left|\sum \sqrt{x(t_k)^2+y(t_k)^2}(t_k-t_{k-1})\right|.$$

Taking limits $\Delta \to 0$ on both sides, we obtain the desired inequality.

17.2.8

Suppose $\sigma: x^2+y^2=R^2$, show that $\lim_{R\to\infty} I_R = 0$ where

$$I_R = \int_\sigma \frac{y\mathrm{d}x - x\mathrm{d}y}{(x^2+y^2+xy)^2}.$$

Proof: Note that $L(\sigma) \leqslant 2\pi R$ and $\frac{y}{(x^2+y^2+xy)^2} \leqslant 4R^{-3}$, so the integral $|I_R| \leqslant \frac{8\pi}{R^2} \to 0$.